

PRODUCT DOCUMENT

SpaceLift

Full Feature Breakdown

September 2026

Key Features

CORE PRODUCT

1 Structured Intake Flow (10 steps)

- Photo upload (JPG, PNG, WebP, HEIC, up to 5MB, optional)
- Room type selection (Living room, Studio, Bedroom, Dining, Kitchen, Office)
- Occasion field (Everyday, Birthday party, Housewarming, etc.)
- Wall width measurement (feet + inches, with a "count doors" helper tip)
- Ceiling height (optional, with quick-pick presets)
- Budget entry with snap points (\$0 through \$500, or custom)
- Budget cap type (Hard cap, Soft cap, No limit)
- Mood selection (Cozy, Festive, Sleek, Minimalist, Romantic, Bright) with ambient/task/accent lighting mix
- Live lighting preview — AI generates an evening-lit version of the uploaded photo during intake
- Lease rules / damage level (No damage, Minor OK, Anything goes)
- Review screen with a summary of all choices before generating

2 AI-Generated Design Image

- Calls an AI edge function to produce a personalized room design based on the uploaded photo and all intake parameters
- Auto-generates on first load if no image exists yet
- Can be regenerated from the workspace when project parameters change
- Before/After toggle to compare the original photo against the AI design

3 Shopping List with Real Items

- Each item has a real store name, price, category (Lighting, Decor, Textiles, Tableware, Furniture), and damage badge
- Damage badge per item: No damage, Minor, or Needs OK (lease-safety filtering)
- Quantity controls with per-item cost calculation
- Lock items to prevent them from being swapped
- Mark items as "I already own this" (removes from budget total)
- Swap drawer with alternative suggestions (name, store, price difference, reason)
- Over-budget detection with a "Find cheaper swaps" prompt
- Filter by lease-safety level (All, No damage, Minor, Needs OK)
- Lease-safe indicator showing whether every item passes the user's lease rules
- Copy list, PDF export, and Share buttons

4 Chat-Based Redesign

- Full chat panel with conversation history saved to the database

- AI can swap items (change name, store, price, wire color)
- AI can add new items or remove existing items
- AI can trigger a full redesign and regenerate the design image
- AI can switch the view between Day and Evening modes
- Quick-prompt suggestions when the chat is empty

5 **Version History**

- Every design change creates a version with a label, total cost, summary, and timestamp
- Restore any past version (sets it as current, loads its items)
- Compare action available per version
- Visual indicator showing which version is currently active

6 **Budget Tracking**

- Live budget bar showing spent vs. total with remaining or over-amount
- Hard cap enforcement (design stays under budget)
- Over-budget warning banner with cheaper-swap suggestions
- Budget reflected across the workspace, shopping list, and intake

7 **Interactive Pins on the Design**

- Numbered pins overlaid on the AI design image, each linked to a shopping list item
- Click a pin to see a floating detail card (item name, store, price, Swap/Lock actions)
- Show/hide pins toggle
- Evening overlay (warm gradient) applied on top of the image

8 **Day / Evening View Toggle**

- Switch the room render between daylight and evening lighting
- Affects both the SVG fallback render and the overlay on the AI image

9 **Compare Three Plans Page**

- Side-by-side comparison of the same room at three budget levels (Budget-friendly, Mid-range, Premium)
- Each plan shows its own mood, lease rules, item count, budget bar, and lighting mix
- Budget ladder visualization at the bottom
- Select a plan to continue with

10 **Setup Plan**

- Step-by-step placement guide for each item
- Estimated total setup time
- Spending-by-category breakdown with percentage bars
- Distinguishes items you own (just place) vs. items to buy

11 Inline Brief Editing in the Workspace

- Project brief pills at the top of the workspace (occasion, room type, budget, mood, lease rules, wall width)
- Click any pill to edit that parameter inline without leaving the workspace
- "Regenerate design" button appears after a parameter is changed

Key Features Summary

The key features form the core product loop — take structured inputs through the intake flow, generate an AI design image, produce a real shopping list with lease-safe items, and let users iterate via chat and version history. Budget tracking, interactive pins, day/evening views, compare plans, setup planning, and inline editing all reinforce that loop.

Secondary Features

SUPPORTING INFRASTRUCTURE

1 Landing Page / Marketing Homepage

- Hero section with animated mood-cycling room preview
- "How it works" three-step explainer
- Feature grid highlighting all capabilities
- Call-to-action buttons for starting a design or signing in

2 User Authentication

- Sign up and sign in with email and password
- Show/hide password toggle
- Error handling and validation (minimum 6 characters)
- Session persistence (stays signed in between visits)
- Sign out from the dashboard or workspace

3 Dashboard / Project Home

- Grid of saved projects with thumbnail previews (uploaded photo or SVG fallback)
- Project status badges (Active, Draft, Archived)
- Project cards show room type, mood, budget, and last-updated date
- "New project" card to start a fresh design
- Opens existing projects into the workspace

4 Database Persistence

- All projects, versions, items, and chat messages saved to Supabase
- Room photos and design images stored in Supabase Storage buckets
- Row-level security so users only see their own projects

5 SVG Room Render Fallback

- Hand-drawn SVG room illustration used when no AI image is available
- Renders different furniture layouts based on selected mood
- Supports day/evening lighting and interactive pins
- Used on the landing page, intake preview, dashboard thumbnails, and compare page

6 Accessibility

- Skip links on every page
- ARIA roles and labels throughout (progress bars, tabs, dialogs, alerts)
- Keyboard navigation support (Enter/Space on buttons, Escape to close)
- Reduced-motion support on the landing page animation

Secondary Features Summary

The secondary features provide the scaffolding around the core loop: getting users in the door (landing page, auth), keeping their work organized (dashboard, persistence), and supporting experiences that don't need AI (SVG fallback, compare page, setup plan). Accessibility runs throughout as a cross-cutting concern.